

General Science





About the Course

Learners engage with the chemical nature of Things, meteorology, space exploration, and robotics as they practice beginning lab skills and expand their use of graphs. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 5

About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

Science: Grade 5

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just “feels right.” Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

General Science: Grade 5

For fifth-grade students or possibly hungry fourth-graders or sixth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
5	General Science: Grade 5 1 time/week 20 min	Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt The Hive Detectives: Chronicle of a Honey Bee Catastrophe The Tornado Scientist

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 5				
Natural History: Grade 5		General Science: Grade 5 Nature Walks & Scouting: Grades 1-8		Labs: Grade 5 Nature Notebook: Grade 5



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 5

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 5

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 5

- ☐ Term 1: a beekeeper
- ☐ Term 2: a meteorologist or weather station
- ☐ Term 3: evening walks, a planetarium

Term Prep & Teacher Tips

Science: Grade 5

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

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- ☐ Term 1: bees p.384-395
- ☐ Term 2: climate and weather p.780-783, the winds and storms p.791-799
- ☐ Term 3: stars p.815-821

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- honey
- food items suggested
- spoons
- small saucepan
- plant fertilizer of your choice (only a small amount)
- water
- flowers (Term 1)
- a strainer or sieve
- 1 c dry soil
- kitchen scale
- kitchen thermometer, any stick-type tool that can be placed into hot water and soil
- oven
- oven mitt
- clock
- chair or stepladder
- stopwatch, such as found on most electronic devices
- freezer
- stove
- 1-2 c white vinegar
- 1/2-1 c baking soda
- flat surface outside to launch rocket
- 10-20' height reference, such as the side of a building, a long piece of wood, or a flagpole
- various supplies by student design
- printables
- optional: barometer
- optional: computer

Reminders

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☐ Note that Lab in week 1 begins with 5 "unknown" food substances (1- 2 Tbsp) to include honey and any other various items, e.g. peanut butter, whole grain bread, white bread, raw egg white, cereal, applesauce, jelly, whole milk, milk substitute, juice, flavored yogurt, plain yogurt, etc. These will be provided to students in 5 similar-looking containers for study. Refer to the lab book for more details.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

General Science: Grade 5



Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt



The Hive Detectives: Chronicle of a Honey Bee Catastrophe



The Tornado Scientist



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

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Household Items - General Science: Grade 5



Quick Links

Related Course Quick Links

- ∞ [Extra Helpings](#)
- ∞ [Outdoor Work](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Foundations \(See Section 13: Science\)](#)

Click THIS text
or scan the QR
code for links.



SAMPLE

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How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there were any dates that they want to keep for their Book of Centuries.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
 - These can be viewed as alternative ways to engage.
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SAMPLE

General Science: Grade 5

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 20m General Science: Grade 5 - Lesson 1

Visit a Hive

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO

Look at the cover picture and the main title. Have you watched bees before? What did you notice about them? What would you investigate if you were a "hive detective"? Read the subtitle. Do you know what a chronicle is? (a factual record) Do you know what a catastrophe is? (a great disaster) So, this is a story about scientists who are studying a disaster in a beehive to learn how it happened in the hopes that they can prevent it from happening again.

→ READ, NARRATE, & DISCUSS

p.1-3 "Put on your" - "home in Wisconsin."

→ SUPPLEMENTAL

∞ Video Link: Why do Honeybees Love Hexagons? (3:58)

∞ Video Link: Why Do Bees Build Hexagons? (2:34)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 2 20m General Science: Grade 5 - Lesson 2

Know the Bees

Materials: The Hive Detectives

→ INTRO

What did you read last time?

→ READ, NARRATE, & DISCUSS

p.4-9 "Mary begins" - "for their keeper."

- Describe the structure of the hive and the jobs of the bees. Draw or diagram, if helpful.

WEEK 3 20m General Science: Grade 5 - Lesson 3

Beekeeping Cross-Country

Materials: The Hive Detectives

→ INTRO

What did you read last time? Some beekeepers manage thousands of bees. Who would need thousands of bees? Let's find out about what happened when one of these beekeepers visited his hives.

→ READ, NARRATE, & DISCUSS

p.11-14 "On November 12," - "a thankless job."

- Contrast how Mary and Dave keep their bees.
- What would it feel like to come upon the animals you care for to find them gone?

→ SUPPLEMENTAL

∞ Video Link: Honey Bee Shortage (2:34)